

Seneca

SEG800

Digital Signal and Image Processing

Color Fundamentals and Image Processing

Seneca College

Overview

- Color Models (Spaces)
- Pseudo-color Image Processing
- Color Image Processing
- Color Transformations

Color Fundamentals

- Sir Isaac Newton discovered the color spectrum (1666)
- Achromatic light- no color; characterized by intensity only
- Chromatic light
 - Radiance- amount of energy flowing from light source
 - Luminance- amount of energy perceived from a light source by an observer
 - Brightness- subjective descriptor- not measurable.
 - Intensity is the closest measurable proxy for brightness.

Figure 7.1

Color spectrum seen by passing white light through a prism.
(Courtesy of the General Electric Co., Lighting Division.)

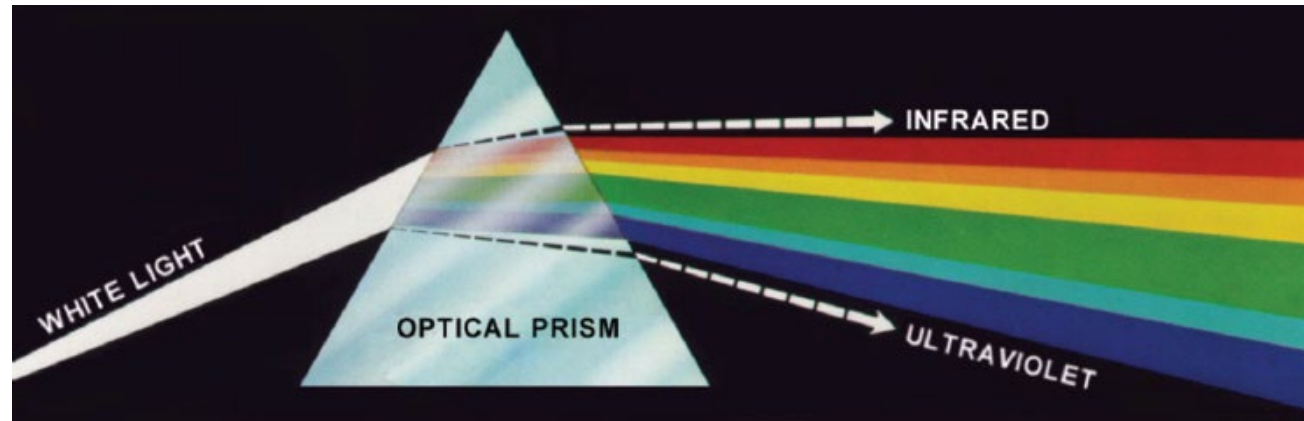
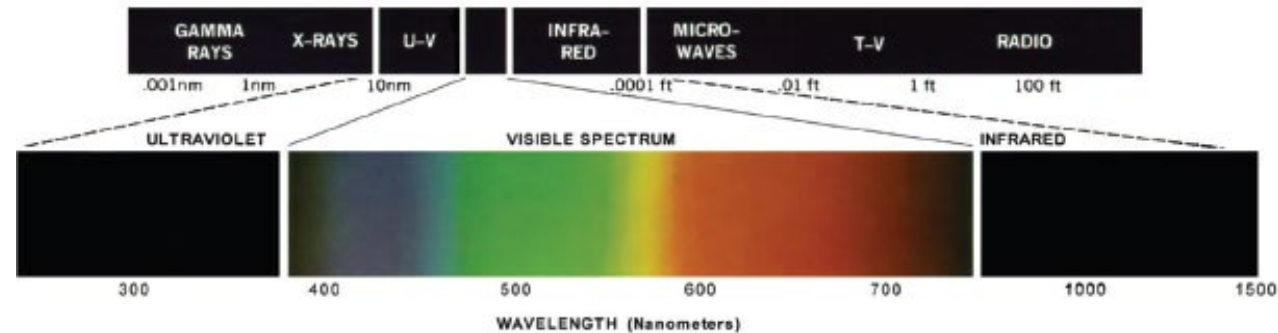


Figure 7.2

Wavelengths comprising the visible range of the electromagnetic spectrum. (Courtesy of the General Electric Co., Lighting Division.)

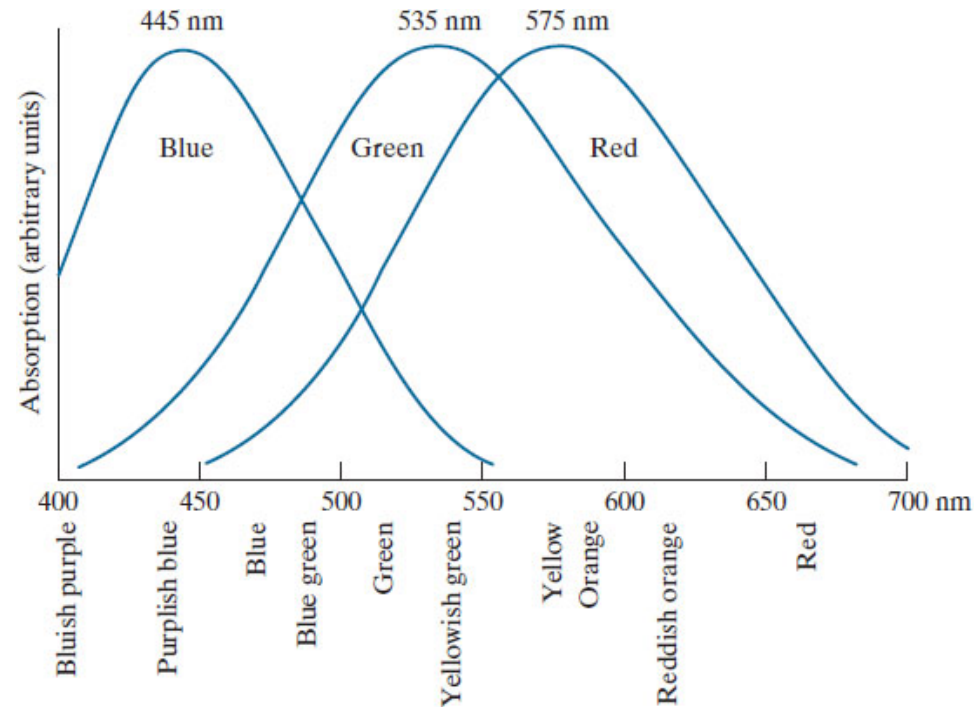


Human Color vision

- Cones
 - 65% sensitive to red
 - 33% sensitive to green
 - 2% sensitive to blue; but most sensitive
- “Because of these absorption characteristics, the human eye sees colors as variable combinations of the so-called primary colors: red (R), green (G), and blue (B).” [DIP]

Figure 7.3

Absorption of light by the red, green, and blue cones in the human eye as a function of wavelength.



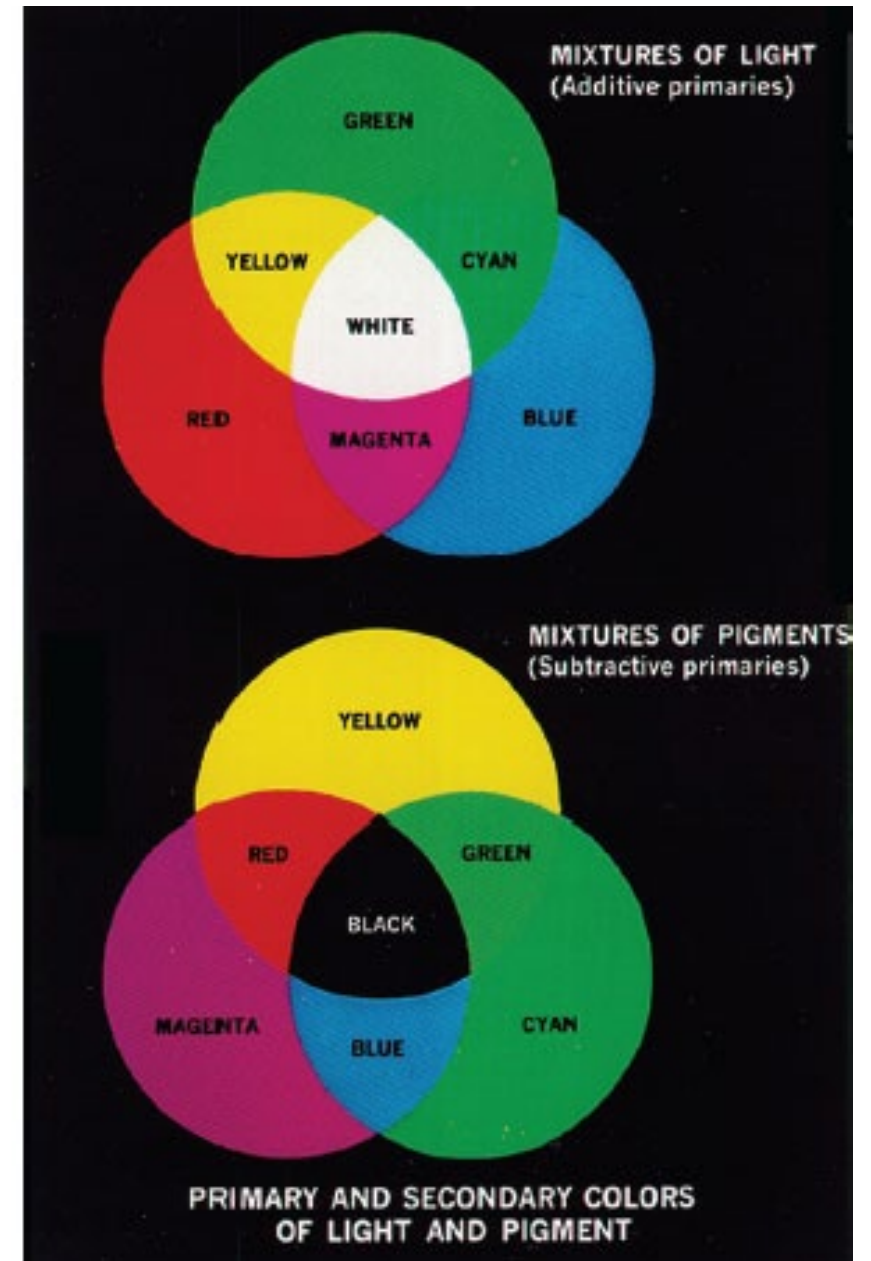
Primary Colors

- Primary colors of light
 - Red, Green, Blue
 - Can be added together to produce secondary colors of light: magenta (red + blue), cyan (green + blue), yellow (red + green)
- Primary colors of pigment
 - Defined as one that absorbs a primary color of light (and reflects the others).
 - Magenta, cyan, yellow

Figure 7.4

Primary and secondary colors of light and pigments.
(Courtesy of the General Electric Co., Lighting Division.)

a
b



Chromaticity

- Tristimulus values: The amounts of red, green, and blue needed to form any color, denoted as X , Y , and Z , respectively.
- Trichromatic coefficients:

$$x = \frac{X}{X + Y + Z}$$

$$y = \frac{Y}{X + Y + Z}$$

$$z = \frac{Z}{X + Y + Z}$$

$$x + y + z = 1$$

Figure 7.5

The CIE chromaticity diagram.
(Courtesy of the General Electric Co.,
Lighting Division.)

- Pure colours on the boundary (spectrum colours)
- Point of equal energy (equal fraction of RGB colours)
- A line segment joining two points: all colour variations obtained by combining the two colours. Similarly for a triangle of any three colours.
- Not all colors can be obtained with three single, fixed primaries (cannot enclose within a triangle). Use of the word primary has been widely misinterpreted.

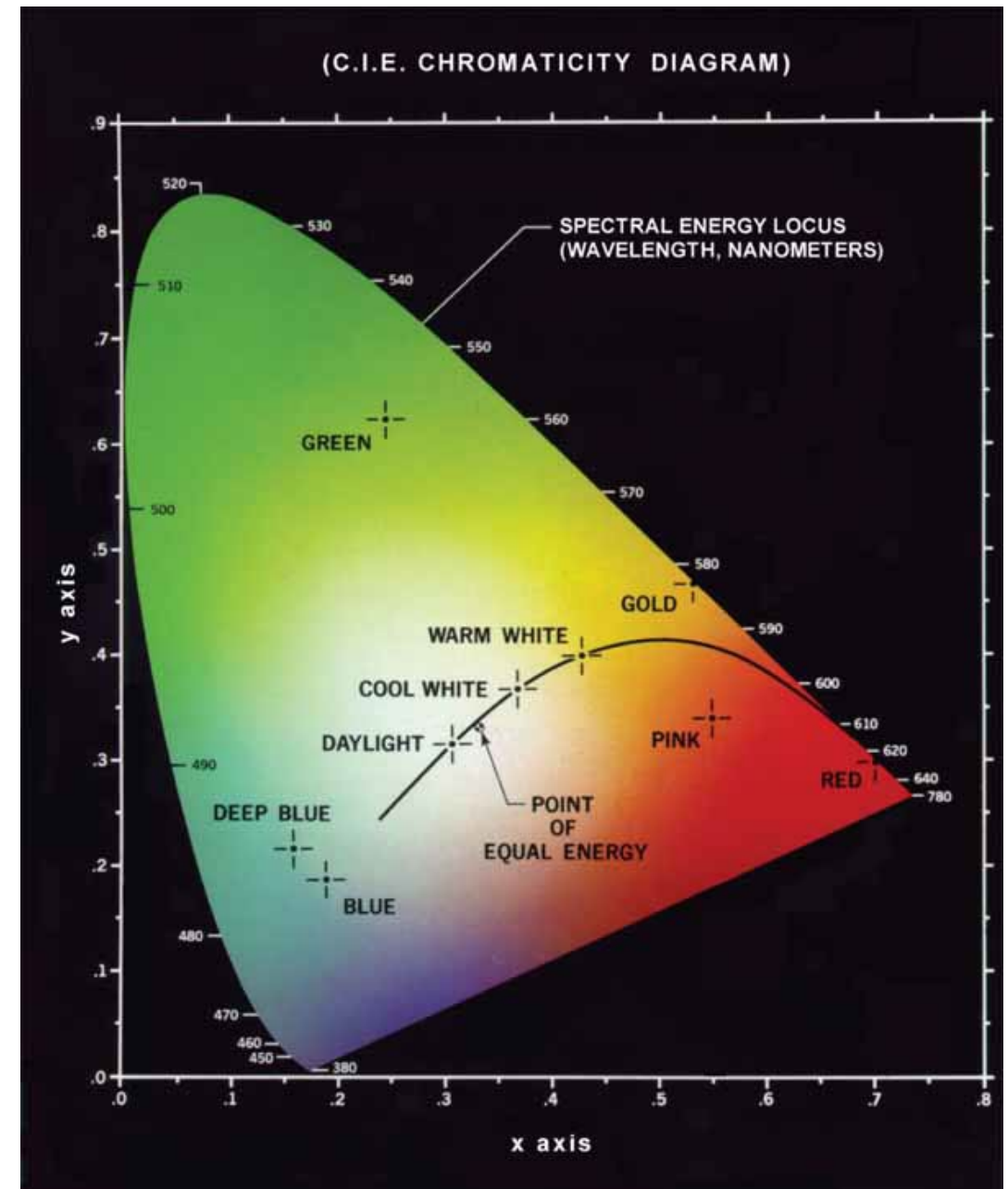
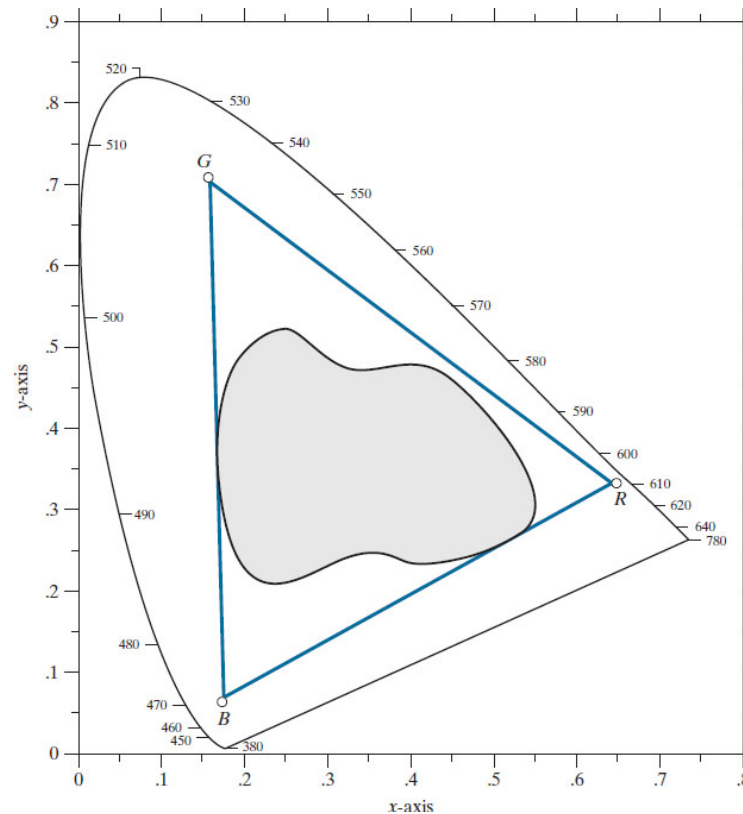


Figure 7.6

Illustrative color **gamut** of color monitors (triangle) and color printing devices (shaded region).



Color Models (or Spaces)

- Suitable for image color generation
 - e.g., capture by camera or display on monitor
 - RGB Color space
- Suitable for Printers
 - CMY Color Space
 - CMYK Color Space
- Suitable for Human manipulation/ creation
 - HSI Color Space
 - $L^*a^*b^*$ Color Space

Figure 7.7

Schematic of the RGB color cube. Points along the main diagonal have gray values, from black at the origin to white at point $(1, 1, 1)$.

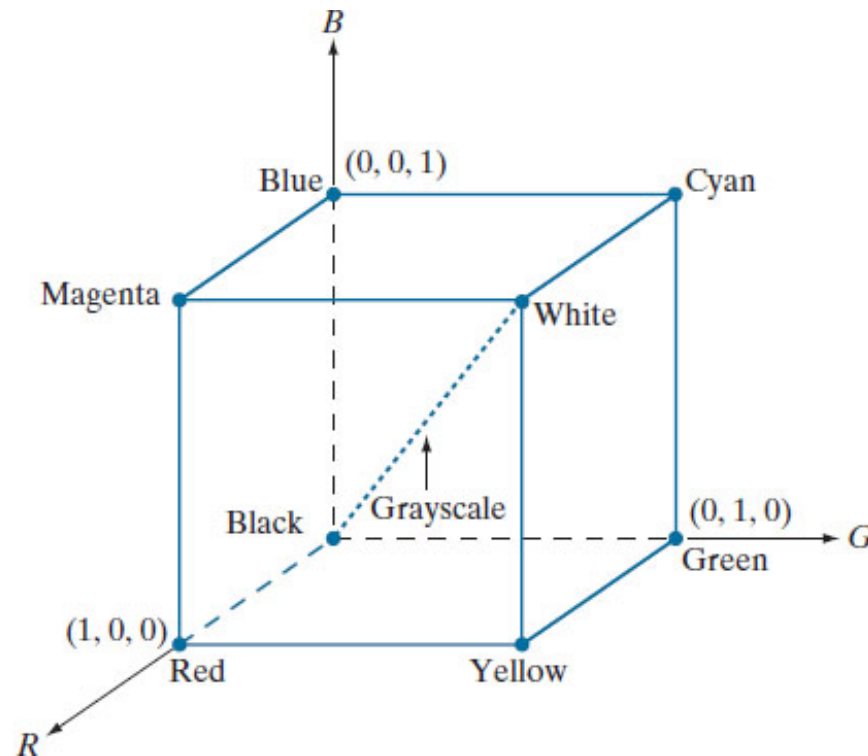


Figure 7.8

A 24-bit RGB color cube.

- Pixel depth: 8 bits (per colour channel).
- Each component 0 to 255
- Total number of possible colours:

$$(2^8)^3 = 16,777,216$$

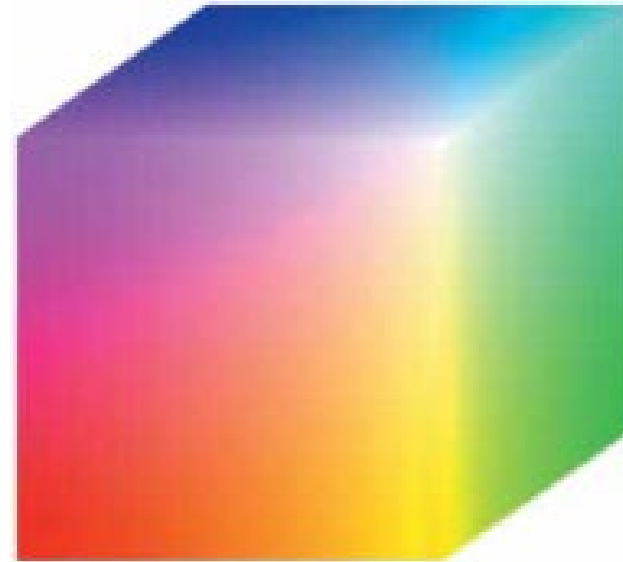
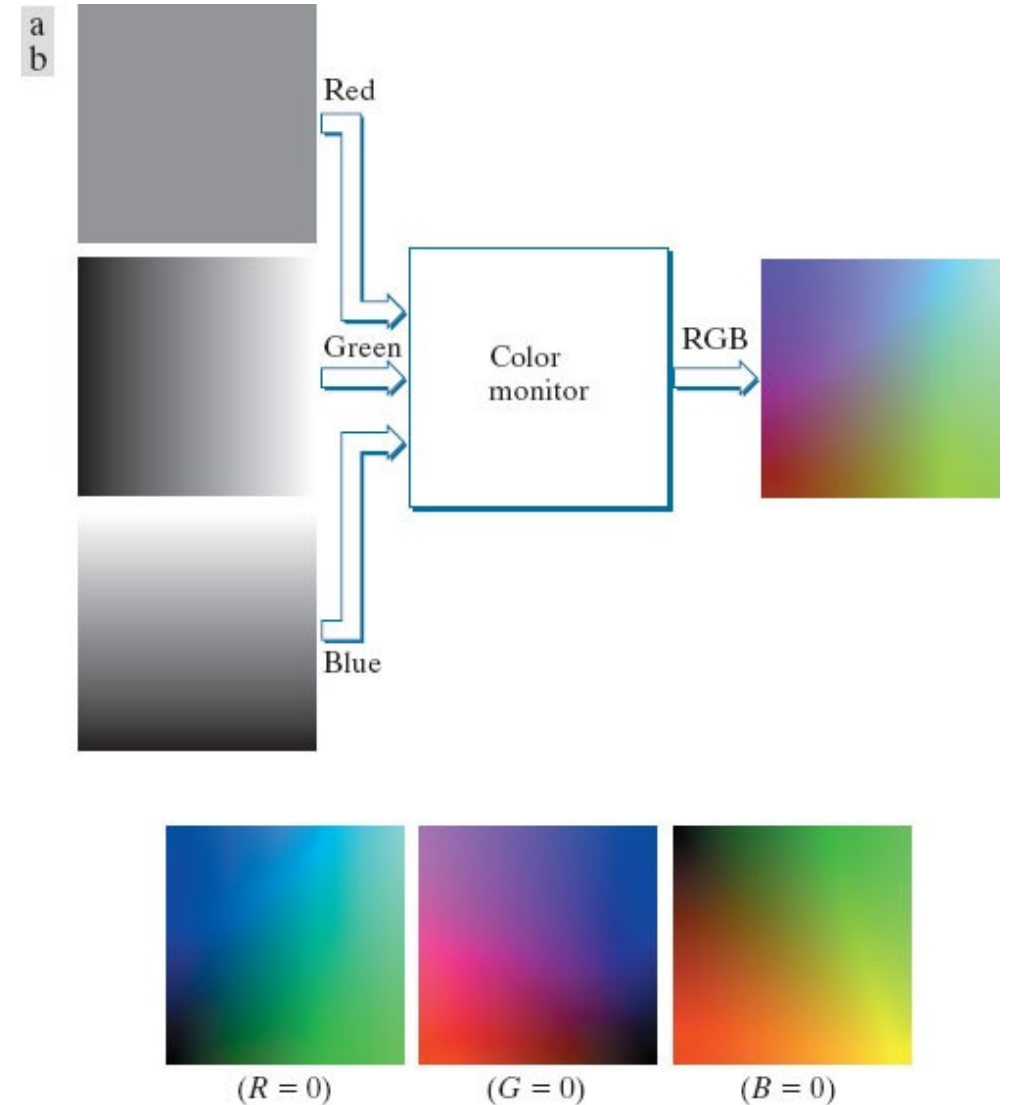


Figure 7.9

(a) Generating the RGB image of the cross-sectional color plane (127, G, B). (b) The three hidden surface planes in the color cube of Fig. 7.8.

- Acquiring a color image is the process shown in Fig. 7.9(a) in reverse. A color image can be acquired by using three filters (sensors), sensitive to red, green, and blue, respectively.



CMY Color Space

- Primary colors of pigment

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

- Since the inks are not pure colors, combining them does not really give us black.

CMYK Color Space

- Adding black for better 'black' prints

$$K = \min(C, M, Y)$$

- If $K = 1$; update C , M , and Y to 0
- Otherwise, update as

$$C = (C - K)/(1 - K)$$

$$M = (M - K)/(1 - K)$$

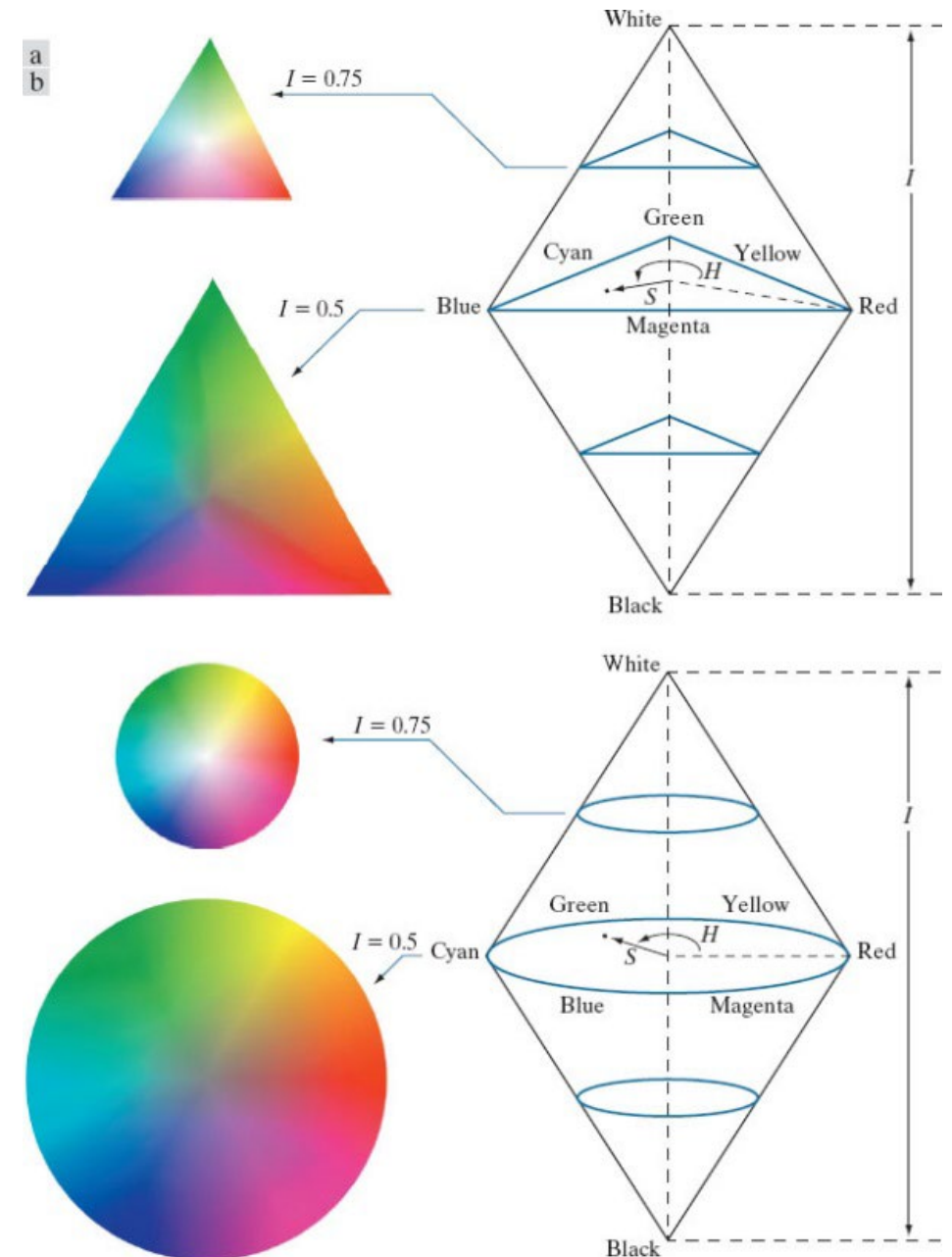
$$Y = (Y - K)/(1 - K)$$

HSI Color Space

- Based on how humans describe color
- Color channels:
 - Hue: describes a pure color, e.g. orange
 - Saturation: how much a pure color is diluted by white.
 - Intensity: as a measurable proxy for brightness, which is subjective and not measurable

Figure 7.12

The HSI color model based on (a) triangular, and (b) circular color planes. The triangles and circles are perpendicular to the vertical intensity axis.



RGB to HSI

$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360 - \theta & \text{if } B > G \end{cases}$$

- Can then be normalized to [0, 1] by dividing by 360

$$\theta = \cos^{-1} \left\{ \frac{\frac{1}{2}[(R - G) + (R - B)]}{\left[(R - G)^2 + (R - B)(G - B) \right]^{1/2}} \right\}$$

$$S = 1 - \frac{3}{(R + G + B)} [\min(R, G, B)]$$

$$I = \frac{1}{3}(R + G + B)$$

- S and I are normalized if R, G, B are normalized to the range [0, 1]

HSI to RGB

- RG sector ($0 \leq H < 120$)

$$B = I(1 - S) \quad R = I \left[1 + \frac{S \cos H}{\cos(60^\circ - H)} \right] \quad G = 3I - (R + B)$$

- GB sector ($120 \leq H < 240$)

$$H = H - 120^\circ$$
$$R = I(1 - S) \quad G = I \left[1 + \frac{S \cos H}{\cos(60^\circ - H)} \right] \quad B = 3I - (R + G)$$

- BR sector ($240 \leq H < 360$)

$$H = H - 240^\circ$$
$$G = I(1 - S) \quad B = I \left[1 + \frac{S \cos H}{\cos(60^\circ - H)} \right] \quad R = 3I - (G + B)$$

CIE $L^*a^*b^*$

- CIE: Commission Internationale de l'Eclairage- the International Commission on Illumination
- Channels:
 - L^* : Lightness / intensity
 - a^* : red minus green
 - b^* : green minus blue
- Characteristics:
 - Device-independent: color consistency between monitors and prints
 - Colorimetric: Colors perceived as matching are encoded the same.
 - Perceptually uniform: Color differences among various hues are perceived uniformly.
 - The entire visible spectrum

Pseudo-color Image Processing

Pseudocolor Image Processing

- Humans can see (distinguish between)
 - less than two dozen shades of gray
 - But thousands of color shades
- Adding pseudocolor (false or fake color) to grayscale images helps us see details better.
 - Intensity slicing and color coding
 - Use thresholds to divide gray values into two or more intervals. Assign a color to each and color the image accordingly.
 - Intensity to color transformations
 - Apply 3 different transformations to intensity image and combine as RGB values.

Figure 7.16

Graphical interpretation of the intensity slicing technique.

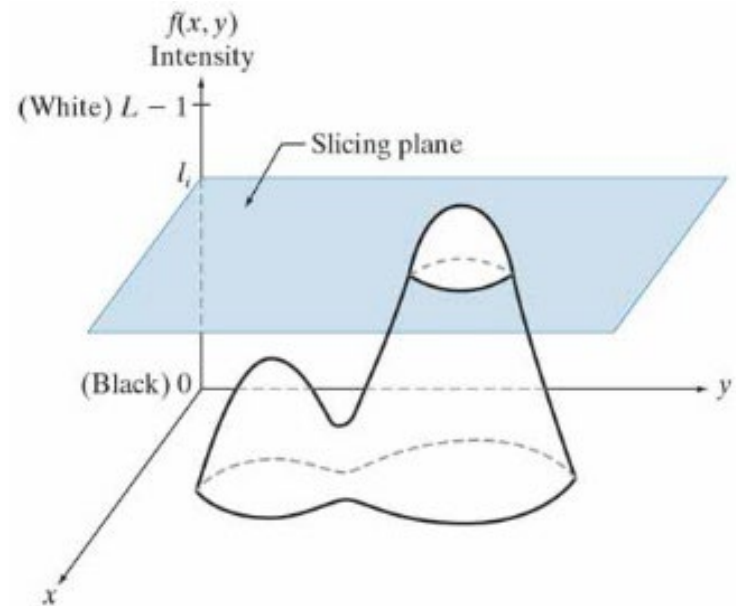


Figure 7.17

An alternative representation of the intensity slicing technique.

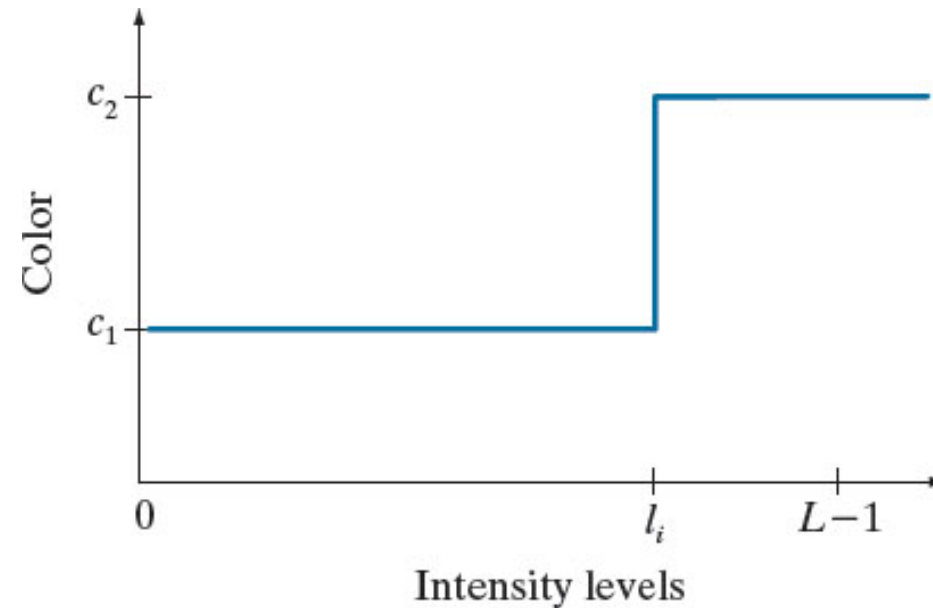


Figure 7.19

(a) X-ray image of a weld. (b) Result of color coding.
(Original image courtesy of X-TEK Systems, Ltd.)

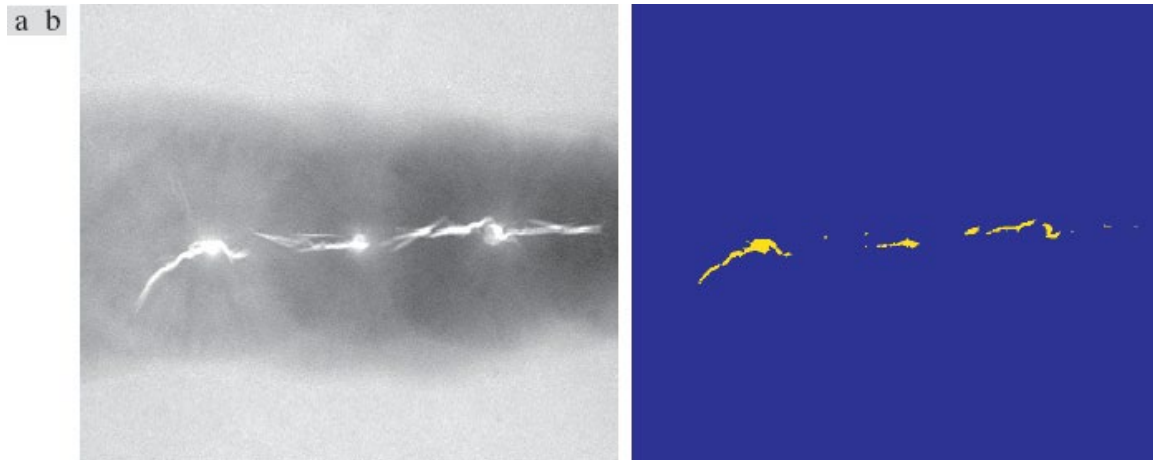


Figure 7.18

(a) Grayscale image of the Picker Thyroid Phantom. (b) Result of intensity slicing using eight colors. (Courtesy of Dr. J. L. Blankenship, Oak Ridge National Laboratory.)

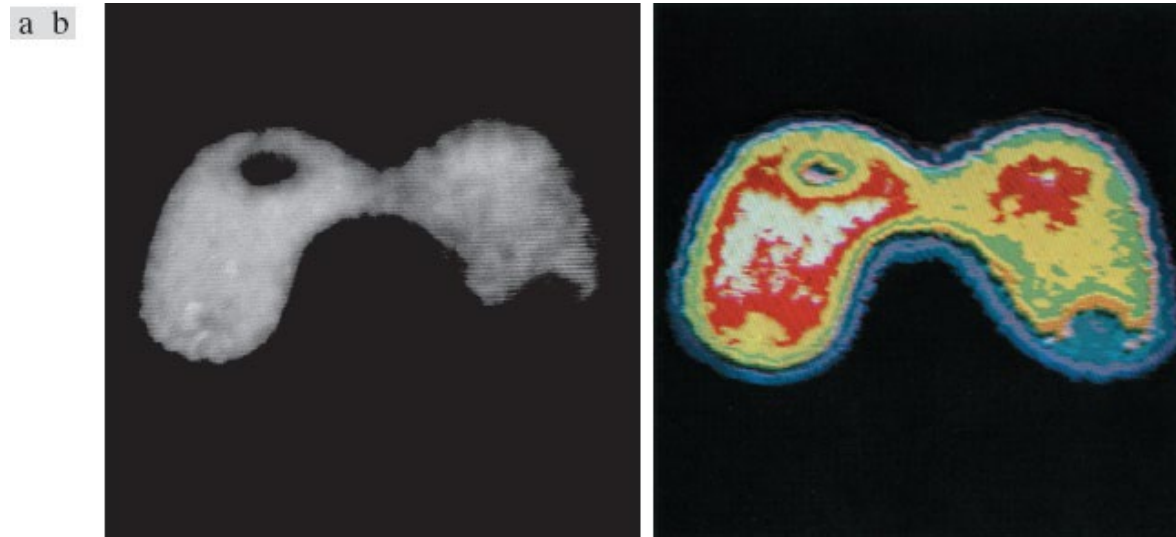
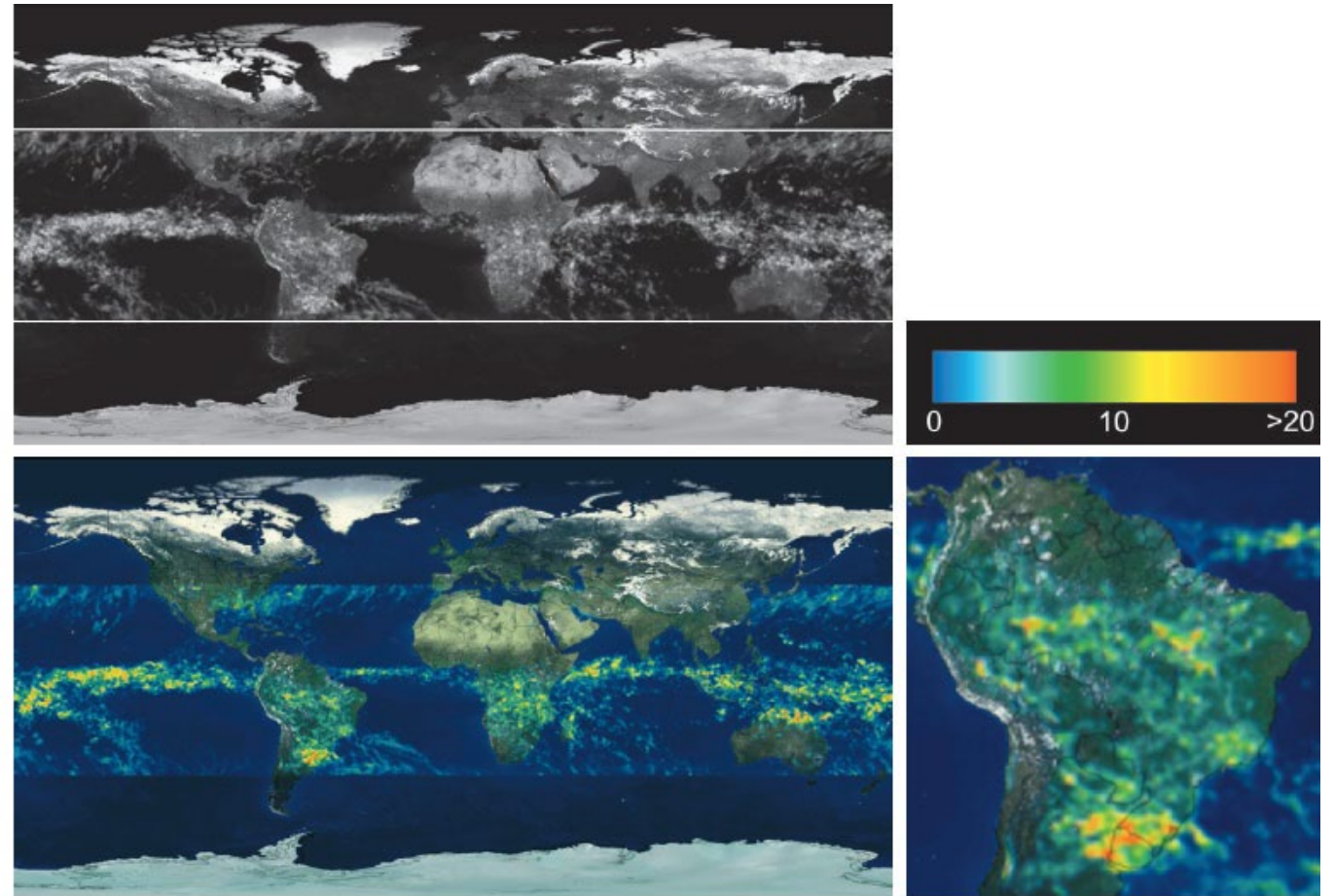


Figure 7.20

(a) Grayscale image in which intensity (in the horizontal band shown) corresponds to average monthly rainfall. (b) Colors assigned to intensity values. (c) Color-coded image. (d) Zoom of the South American region. (Courtesy of NASA.)



a b
c d

Figure 7.21

Functional block diagram for pseudocolor image processing. Images f_R , f_G , and f_B are fed into the corresponding red, green, and blue inputs of an RGB color monitor.

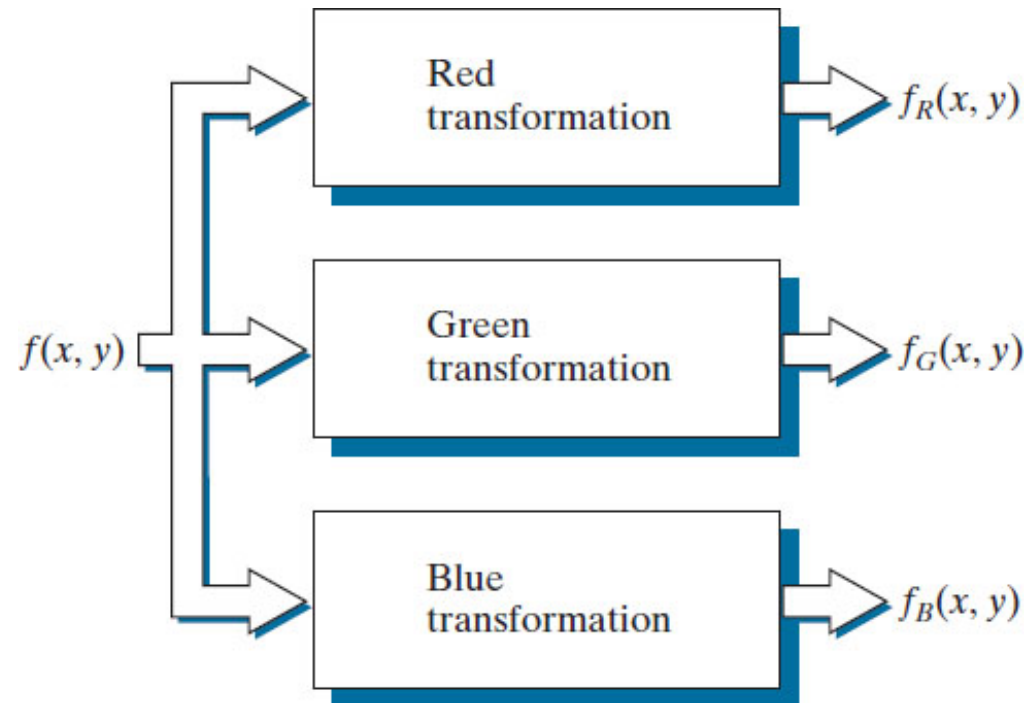


Figure 7.22

Pseudocolor enhancement by using the gray level to color transformations in Fig. 7.23. (Original image courtesy of Dr. Mike Hurwitz, Westinghouse.)

a
b c

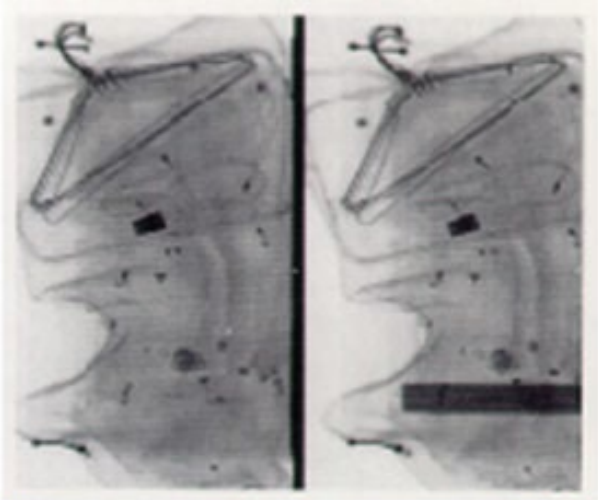


Figure 7.23

Transformation functions used to obtain the pseudocolor images in Fig. 7.22.

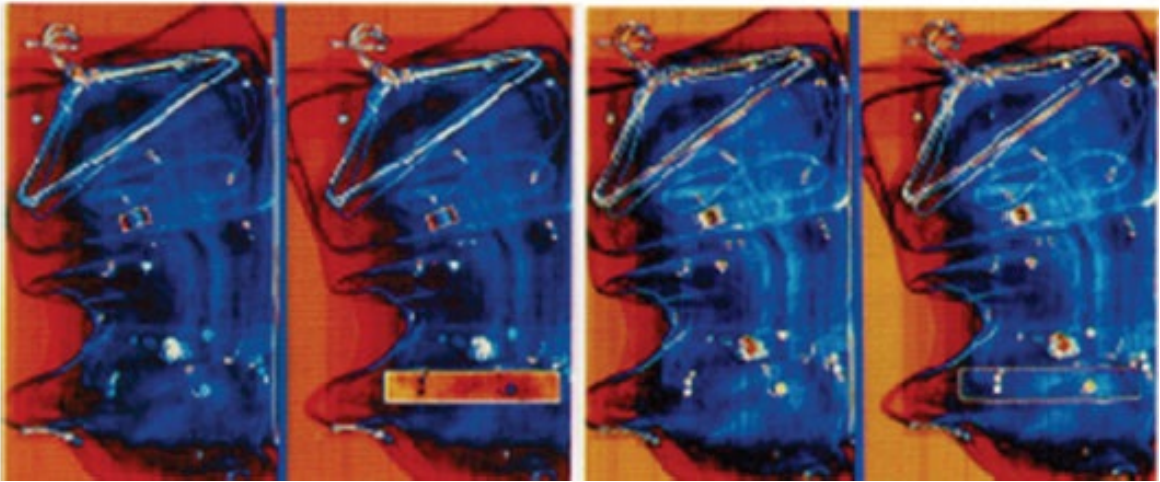
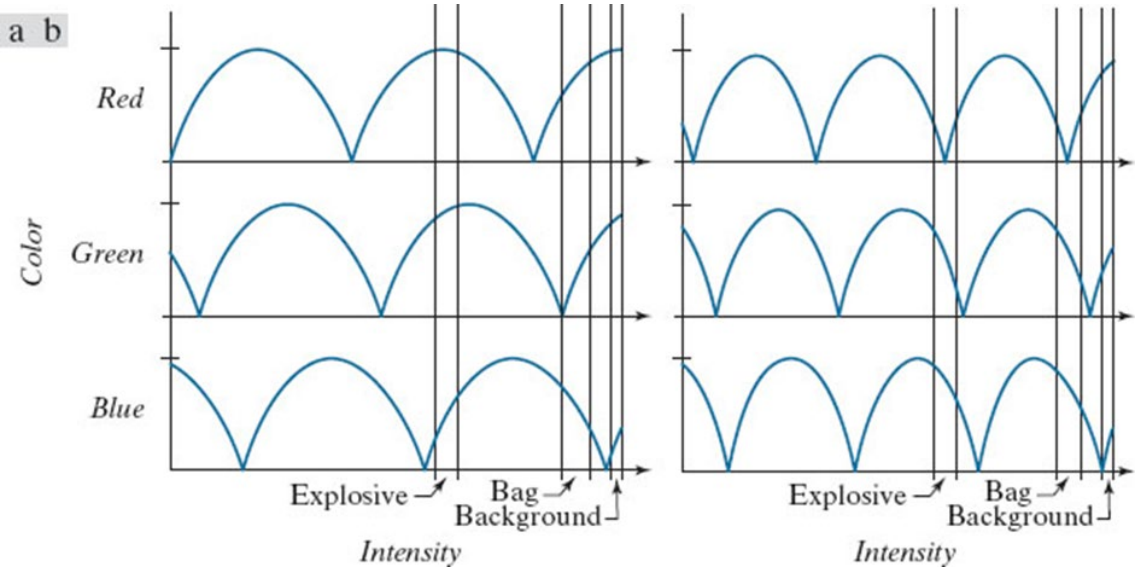


Figure 7.24

A pseudocolor coding approach using multiple grayscale images. The inputs are grayscale images. The outputs are the three components of an RGB composite image.

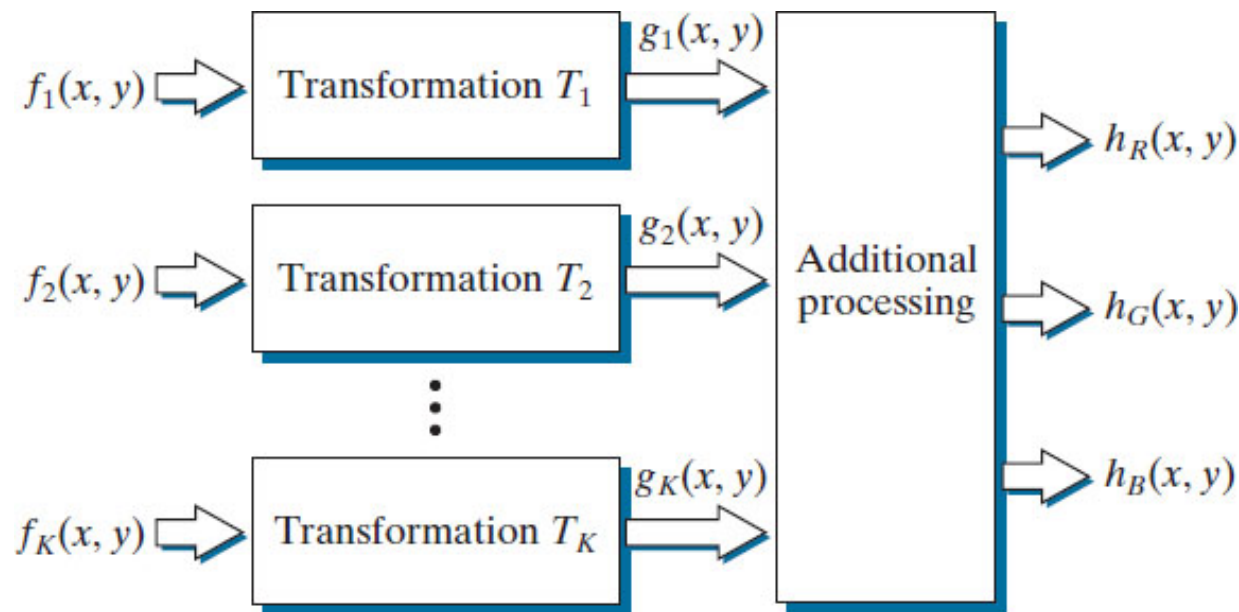


Figure 7.25

(a)–(d) Red (R), green (G), blue (B), and near-infrared (IR) components of a LANDSAT multispectral image of the Washington, D.C. area. (e) RGB color composite image obtained using the IR, G, and B component images. (f) RGB color composite image obtained using the R, IR, and B component images. (Original multispectral images courtesy of NASA.)

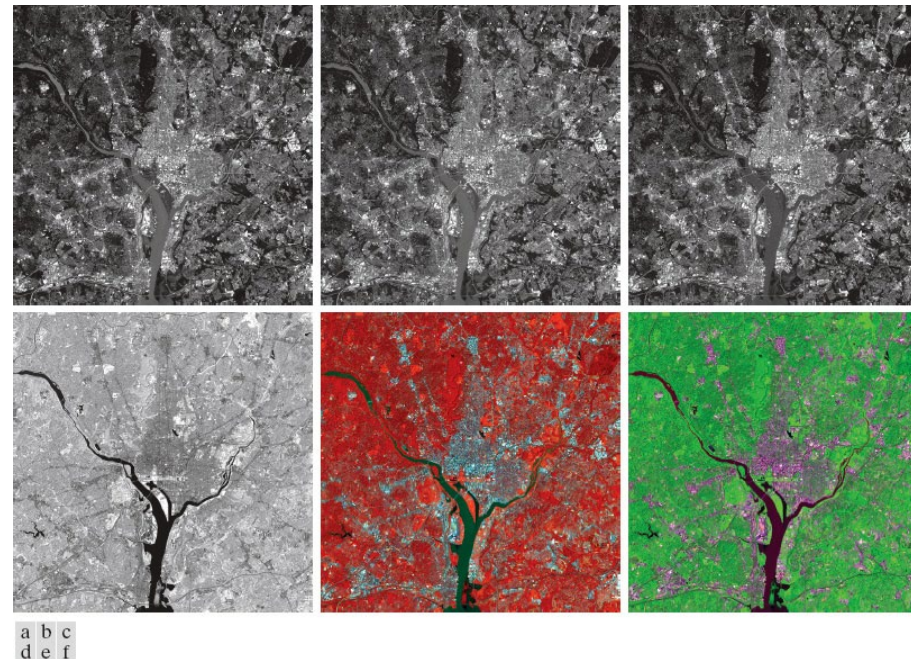
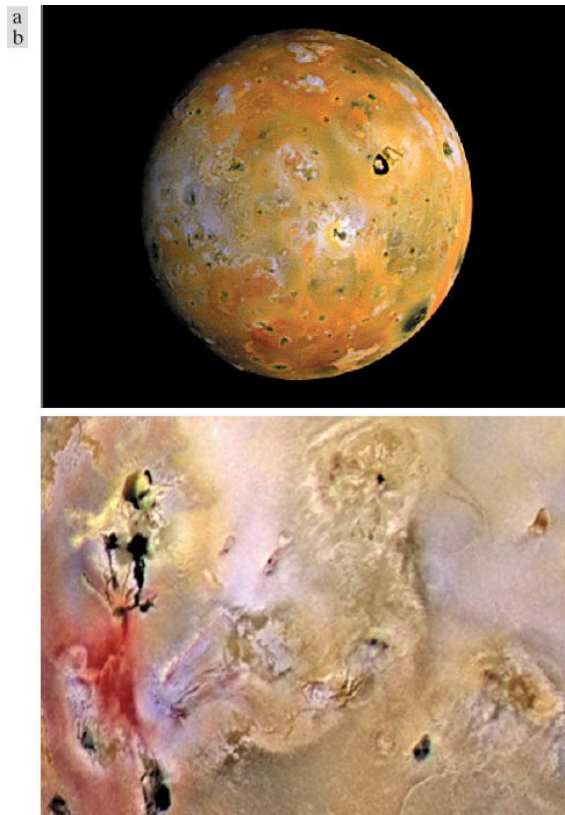


Figure 7.26

(a) Pseudocolor rendition of Jupiter Moon Io. (b) A close-up.
(Courtesy of NASA.)



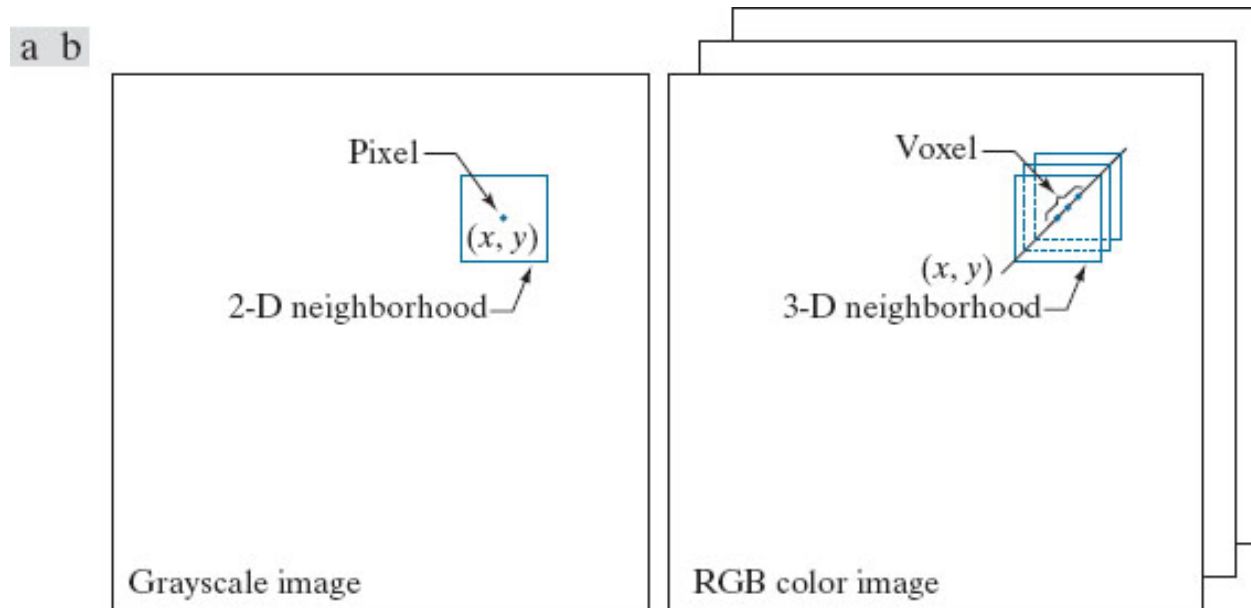
Color Image Processing

Color Image Processing Approaches

- Two major categories:
 - Process each component (color channel) separately, then put back together
 - Process all components directly
 - For example,
 - Color transformation: Convert a color to another.
 - Convert to HSI, work on intensity channel only, convert back

Figure 7.27

Spatial neighborhoods for grayscale and RGB color images. Observe in (b) that a single pair of spatial coordinates, (x,y) , addresses the same spatial location in all three images.



Color Image Averaging/ Smoothing

- Method 1: Process each component (color channel) separately, then put back together

$$\bar{\mathbf{c}}(x, y) = \frac{1}{K} \sum_{(s,t) \in S_{xy}} \mathbf{c}(s, t)$$
$$\bar{\mathbf{c}}(x, y) = \begin{bmatrix} \frac{1}{K} \sum_{(s,t) \in S_{xy}} R(s, t) \\ \frac{1}{K} \sum_{(s,t) \in S_{xy}} G(s, t) \\ \frac{1}{K} \sum_{(s,t) \in S_{xy}} B(s, t) \end{bmatrix}$$

- Method 2: Convert to HSI, smooth the intensity channel only, convert back to RGB

Figure 7.36

(a) RGB image. (b) Red component image. (c) Green component. (d) Blue component.



Figure 7.37

HSI components of the RGB color image in Fig. 7.36(a). (a) Hue. (b) Saturation. (c) Intensity.



Figure 7.38

Image smoothing with a 5×5 averaging kernel. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.



a b c

Figure 7.39

Image sharpening using the Laplacian. (a) Result of processing each RGB channel. (b) Result of processing the HSI intensity component and converting to RGB. (c) Difference between the two results.



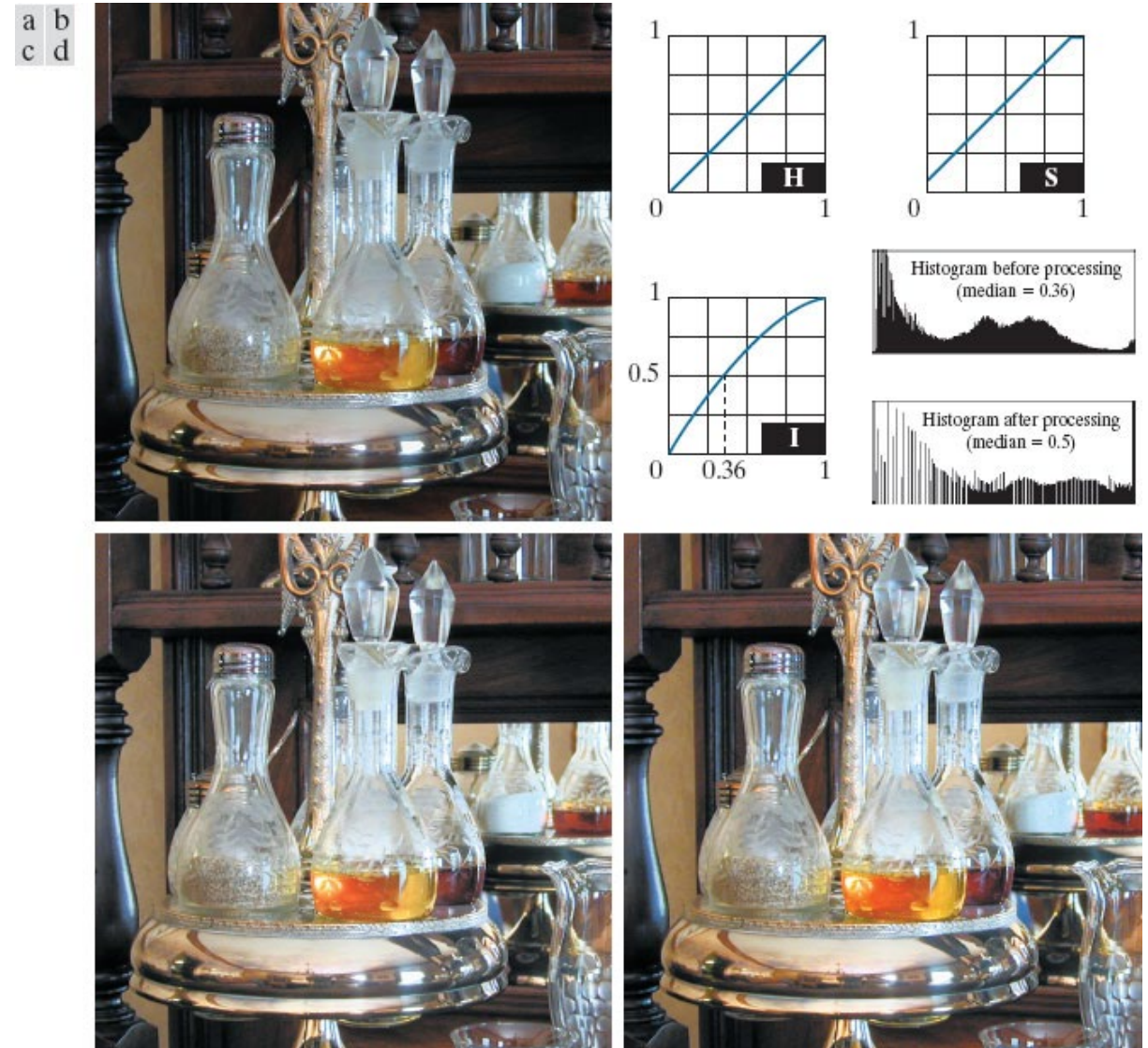
Figure 7.35

Histogram equalization (followed by saturation adjustment) in the HSI color space.

(c) Histogram equalizing the intensity component, without altering the hue and saturation

(d) Increasing the image's saturation component after histogram equalization

Note: Histogram equalization applied individually to each RGB color channel does not make sense.



Color Transformations

Color Transformation

- Transform an intensity value r to intensity level s

$$s_i = T_i(r_i) \quad i = 1, 2, \dots, n$$

- For each color channel ($n=3$ for 3 color channels such as RGB)

- Example: Modify intensity by constant k

- In HSI space: $s_3 = kr_3$

- In RGB space: $s_i = kr_i \quad i = 1, 2, 3$

- In CMY space: $s_i = kr_i + (1 - k) \quad i = 1, 2, 3$

- In CMYK space: $s_i = \begin{cases} r_i & i = 1, 2, 3 \\ kr_i + (1 - k) & i = 4 \end{cases}$

Figure 7.28

A full-color image and its various color-space components. (Original image courtesy of MedData Interactive.)



Figure 7.29

Adjusting the intensity of an image using color transformations. (a) Original image. (b) Result of decreasing its intensity by 30% (i.e., letting $k = 0.7$). (c) The required RGB mapping function. (d)–(e) The required CMYK mapping functions. (f) The required CMY mapping function. (g)–(h) The required HSI mapping functions. (Original image courtesy of MedData Interactive.)

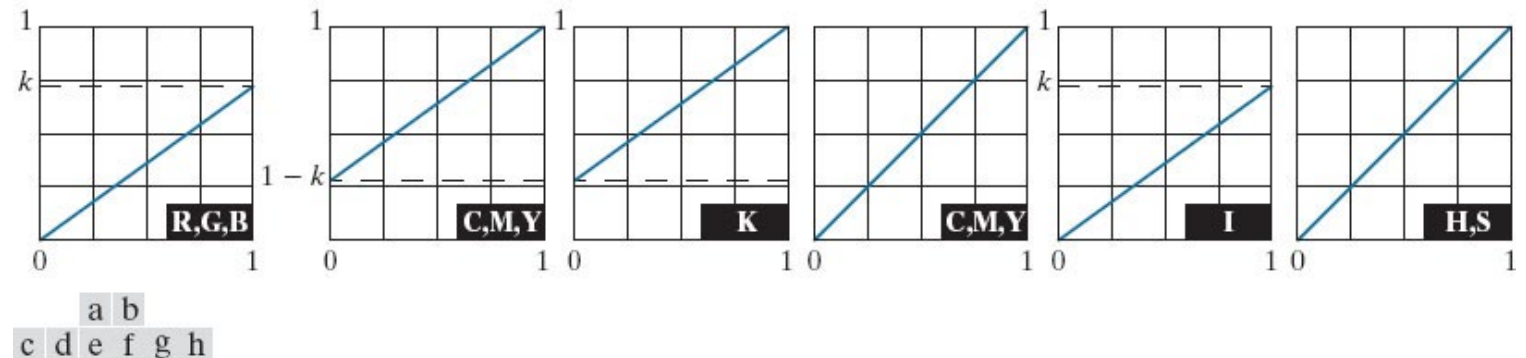


Figure 7.30

Color complements on the color circle.

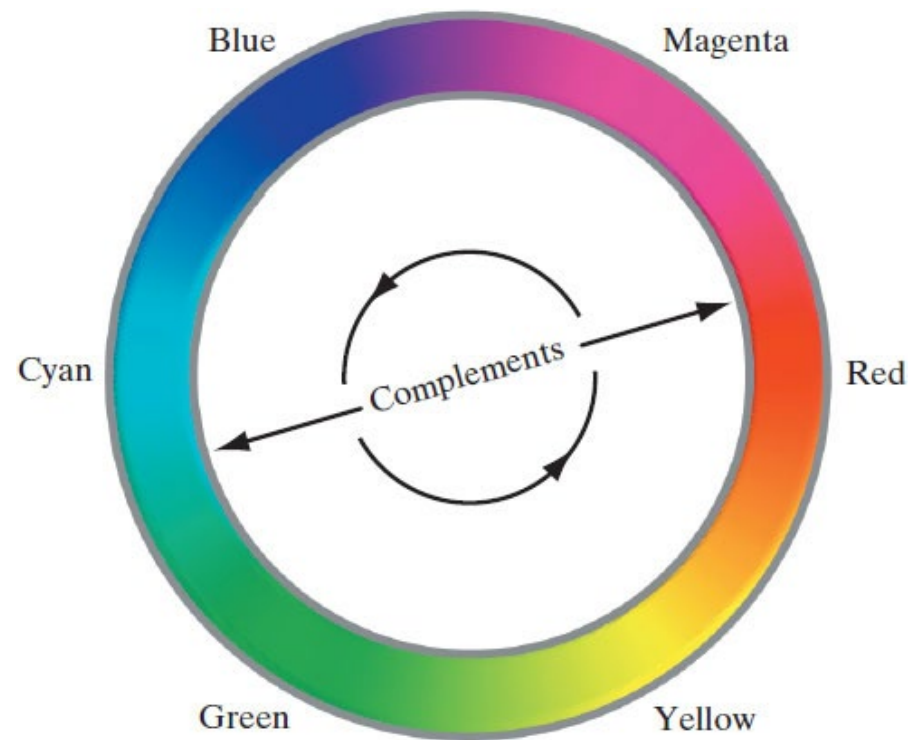


Figure 7.31

Color complement transformations. (a) Original image. (b) Complement transformation functions. (c) Complement of (a) based on the RGB mapping functions. (d) An approximation of the RGB complement using HSI transformations.

a b
c d

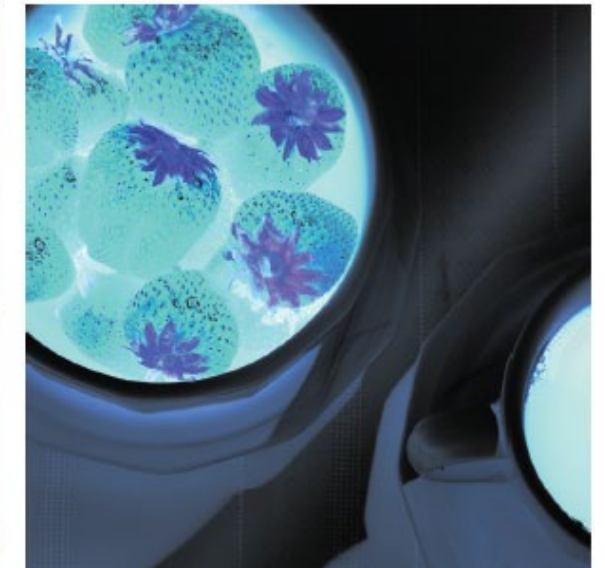
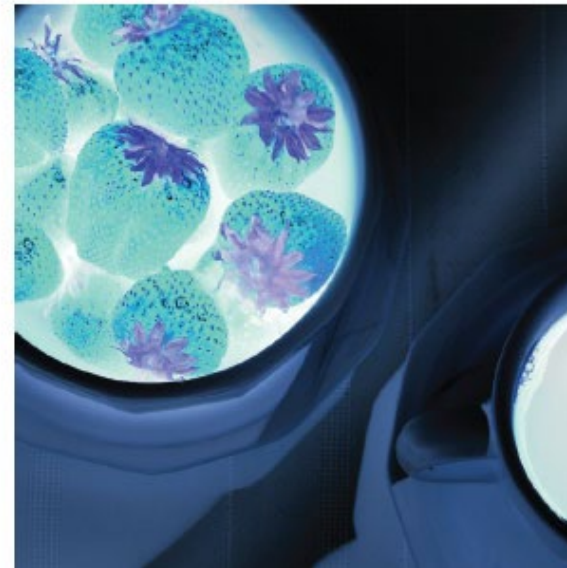
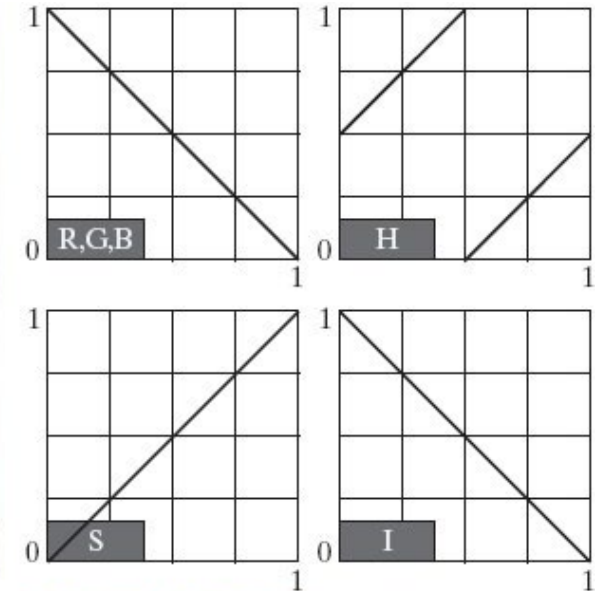
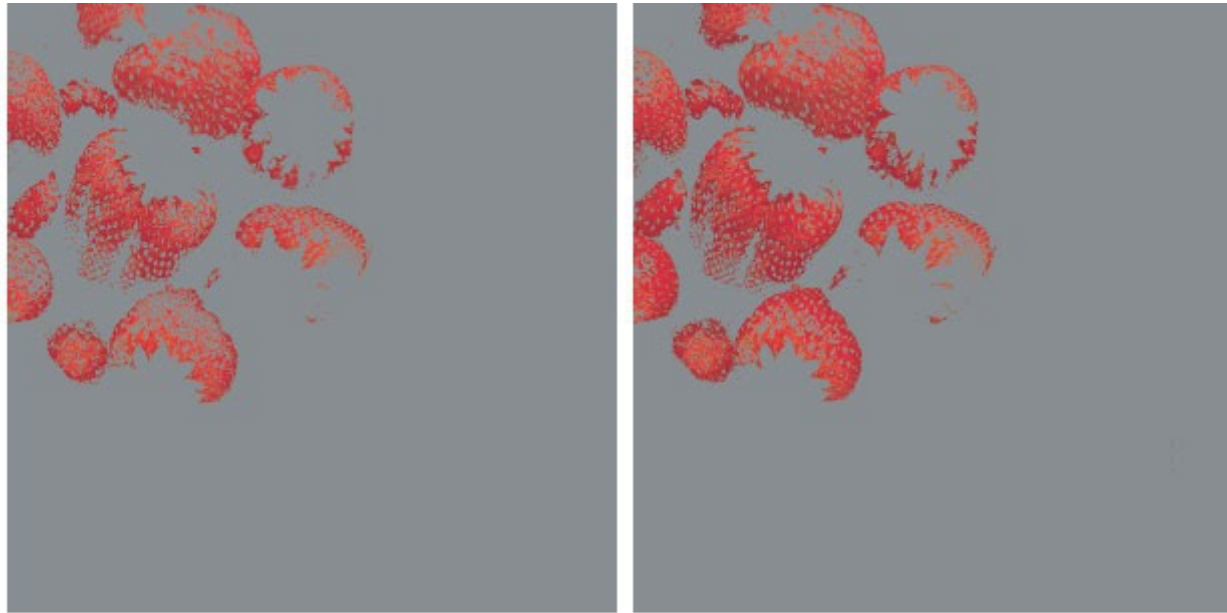


Figure 7.32

Color-slicing transformations that detect (a) reds within an RGB cube of width $W = 0.2549$ centered at $(0.6863, 0.1608, 0.1922)$, and (b) reds within an RGB sphere of radius 0.1765 centered at the same point. Pixels outside the cube and sphere were replaced by color $(0.5, 0.5, 0.5)$.



a b

Figure 7.33

Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not always alter the image hues significantly.

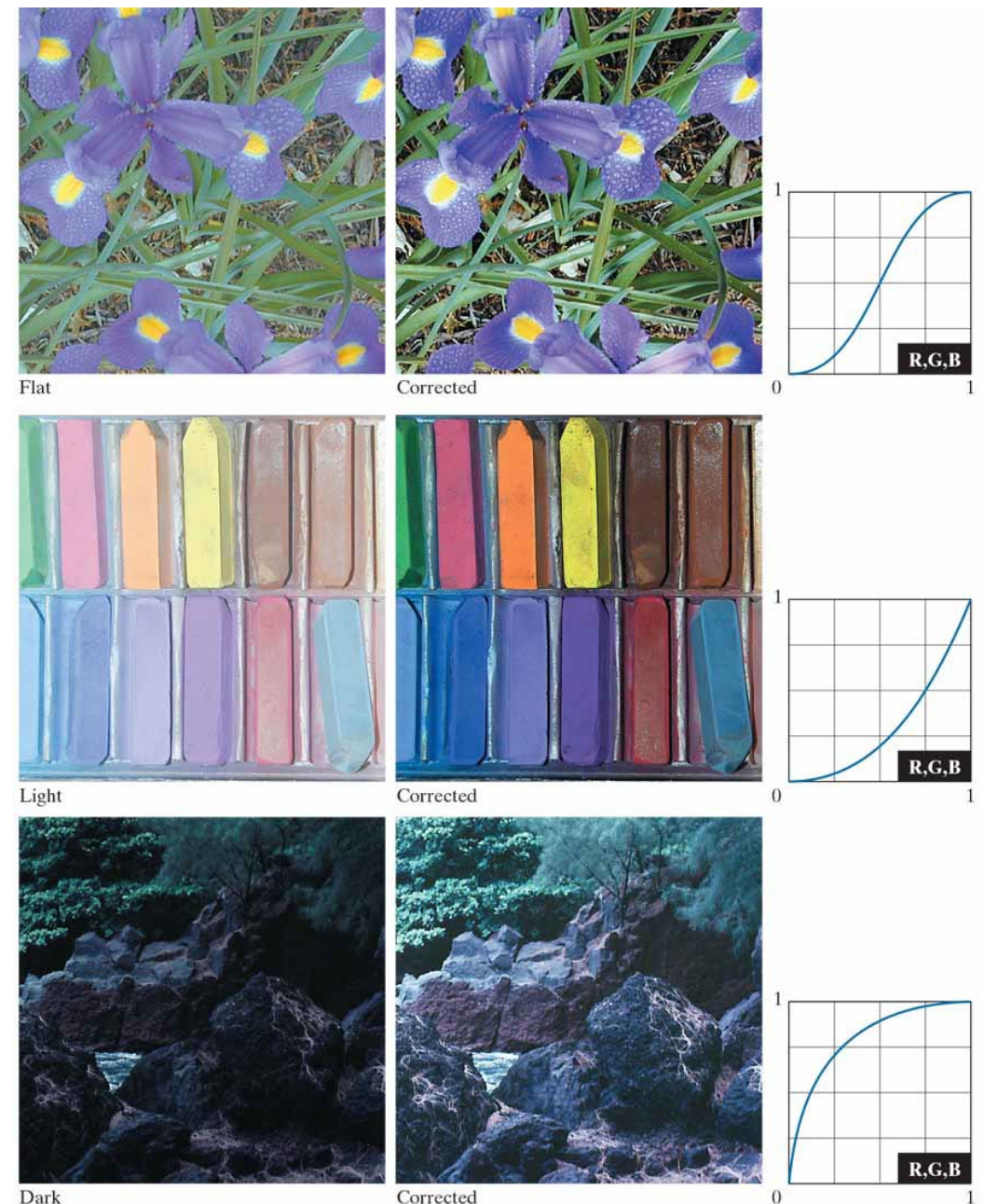
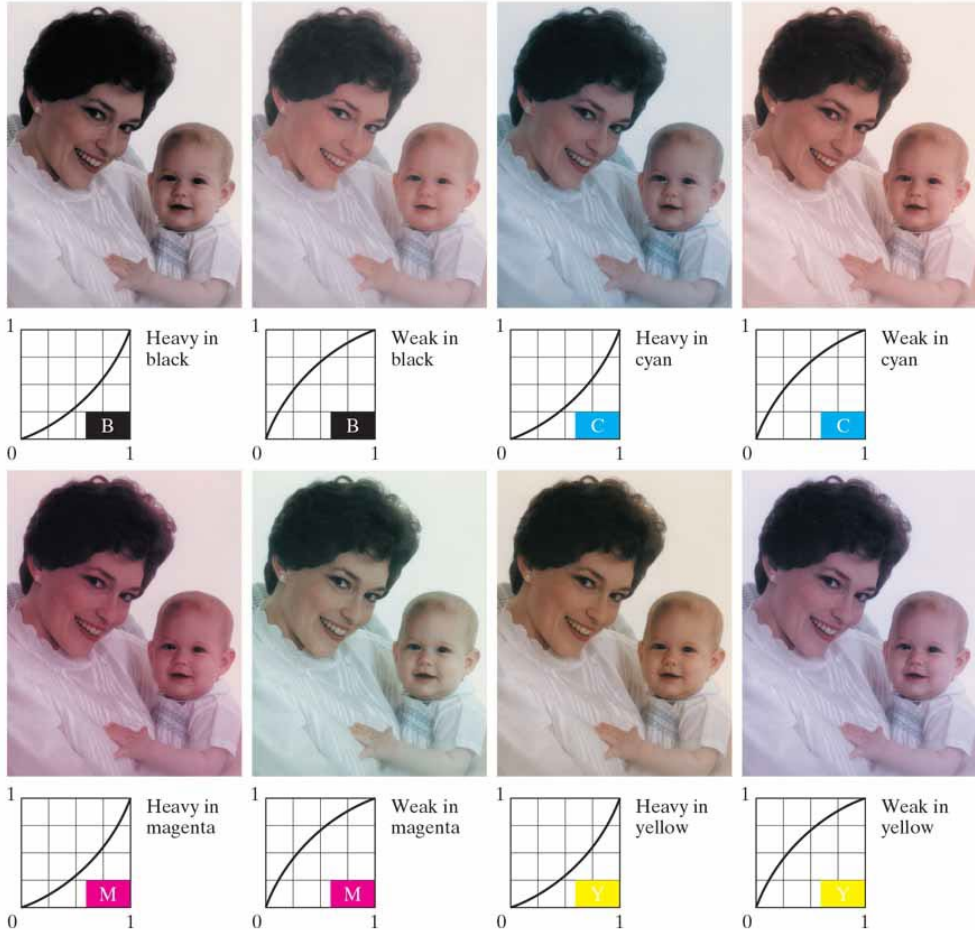


Figure 7.34

Color balancing a CMYK image.



Original/Corrected



Summary

- The human eye sees colors as variable combinations of the so-called primary colors: red (R), green (G), and blue (B), due to cones sensitive to these 3 wavelengths.
- Various color models are suitable for different applications. Examples of color models are: RGB, CMY, CMYK, HSI, and $L^*a^*b^*$.
- Pseudocolor image processing refers to techniques that add fake color to grayscale images to help us see details better.
- Two major approaches to processing color images are : (i) process each color channel separately and then put them back together; and (ii) convert to HSI space, work only on I channel, then convert back to original space.
- Color transformations are applied to enhance images, correct color tones, or balance colors.

References

- [DIP] Gonzalez, R.C. and Woods, R.E., Digital Image Processing, Pearson, 4th ed., 2018.
- [CV] Computer Vision: Algorithms and Applications, R. Szeliski (<http://szeliski.org/Book>)
- [PICV] Practical introduction to Computer Vision with OpenCV, Kenneth Dawson-Howe, WILEY, 2014. Available through Seneca libraries

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